

Visual Defects Are Not Random.

What if surface defects were not inevitable... but predictable?

Naive Bayes · Risk Scoring · Defect Prevention



1,847

Painted Parts

10

Features

76.5%

Accuracy

0.648

ROC-AUC

A surface defect is not just a cosmetic issue — it's a failure you can't hide.



Quality Cost

Scrap, rework, and repaint consume labor and cycle time.



Customer Visibility

A painted trim piece with a scratch goes directly to the customer.



Late Discovery

Defects appear at final inspection but originate upstream.

The defect emerges at inspection — but the root cause lives in injection, storage, and handling. That gap is where ML operates.

Old Thinking

Defects are random events:

"It happened / it didn't."

Inspect at the end.
Reject what fails.
No prevention. No foresight.
Just sorting after the damage.



New Thinking

Defects are probabilistic outcomes:

"How likely is a defect — before painting?"

Score each part before it enters the cabin.
High-risk parts get extra attention.
Low-risk parts flow through.
Defect risk becomes a process input.

This dataset spans four process stages — each contributing its own defect risk:

INJECTION	Regrind %, resin temperature, cooling time <i>Injection molding stage — material and process conditions</i>
PAINT CABIN	Paint viscosity, film thickness, booth humidity <i>Paint application stage — coating quality parameters</i>
LOGISTICS	Storage time, handlings, container type, protection <i>Internal logistics — the dominant defect risk stage</i>
INSPECTION	surface_defect (0 = pass, 1 = defect) <i>Final visual inspection — the binary target variable</i>

Key insight: defects originate in logistics, not in painting. The model quantifies what intuition suspected.

In a real environment, this data lives in four different systems — assembling it is half the work.

A probabilistic model that reasons like an engineer: assign a risk score based on known conditions.

No Scaling Needed

GaussianNB estimates mean and variance per class directly — no StandardScaler required. Simpler, faster, production-ready.

Probabilistic Output

Returns defect probability (0–1), not just pass/fail. Enables risk ranking and threshold adjustment at the operator's discretion.

Pipeline-Ready

sklearn Pipeline bundles preprocessing and classification. The simulator in Section 10 is portable with a single `predict_proba()` call.

Effect Size Auditable

Cohen's d and Mutual Information reveal which features actually separate defect from pass — no black box, full auditability.

76.5%

Accuracy

0.648

ROC-AUC

0.155

F1 (Defect)

21.6%

Defect Rate

Top Drivers — Cohen's d & Mutual Information

pre_paint_storage_hrs	d = 0.189
num_handlings	d = 0.169
part_protection (MI)	Dominant
container_type (MI)	Secondary
resin_temp_c	d = 0.049
regrind_pct	d = 0.002

Confusion Matrix (370 test parts)

	Pred: Pass	Pred: Defect
Actual: Pass	275 (TN)	15 (FP)
Actual: Defect	72 (FN)	8 (TP)

Low recall at 0.5 threshold — the model's real value is the probability score, not the binary verdict.

Storage Time $d=0.189$

→ Cap pre-paint waiting at 4 hours. Parts stored longer show measurably higher defect rates — contamination and static accumulate.

Handlings $d=0.178$

→ Reduce repositioning steps in internal logistics. Each extra handling = +0.3pp defect risk on average.

Unprotected 29.3% rate

→ Make part protection mandatory for all batches entering the paint queue. Protected parts show 15.9% vs 29.3% defect rate.

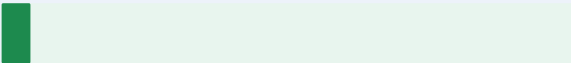
Cardboard 29.5% rate

→ Migrate from cardboard pallets to plastic boxes. Plastic_box containers show 15.1% vs 29.5% for cardboard.

Injection params

→ Resin temp, cooling time, regrind have near-zero Cohen's d — the defect is not born at injection. Focus upstream effort on logistics.

Same model. Three scenarios. The third one isolates the value of a single operational decision.

Scenario A — Low Risk		Scenario B — High Risk		Scenario C — Intervention	
Storage	1 hr	Storage	18 hrs	Storage	18 hrs
Handlings	2	Handlings	6	Handlings	6
Container	plastic_box	Container	metal_rack	Container	metal_rack
Protection	protected	Protection	unprotected	Protection	protected ✓
Temp	235°C	Temp	248°C	Temp	248°C
					
4.6% defect probability		91.9% defect probability		82.7% defect probability	

Scenario C insight: adding protection alone drops the risk by only 9.2pp. The dominant drivers are storage time and handling count — not packaging.

This project is not about finding defects.

It is about preventing them.

Naive Bayes does not replace inspection —
it shifts the decision upstream, before the damage is done.

Feel free to download the full project: dataset, notebook, and simulator.

Change the thresholds. Adjust the scenarios. Challenge the assumptions.

Because quality has no finish line.

